

1. Two Furthest Houses With Different Colors

There are n houses evenly lined up on the street, and each house is beautifully painted. You are given a 0-indexed integer array `colors` of length n , where `colors[i]` represents the color of the i th house. Return the maximum distance between two houses with different colors.

The distance between the i th and j th houses is $\text{abs}(i - j)$, where $\text{abs}(x)$ is the absolute value of x .

1.1. Examples:



Input: `colors = [1,1,1,6,1,1,1]`

Output: 3

Explanation: In the above image, color 1 is blue, and color 6 is red.

The furthest two houses with different colors are house 0 and house 3.

House 0 has color 1, and house 3 has color 6. The distance between them is $\text{abs}(0 - 3) = 3$.

Note that houses 3 and 6 can also produce the optimal answer.



Input: `colors = [1,8,3,8,3]`

Output: 4

Explanation: In the above image, color 1 is blue, color 8 is yellow, and color 3 is green.

The furthest two houses with different colors are house 0 and house 4.

House 0 has color 1, and house 4 has color 3. The distance between them is $\text{abs}(0 - 4) = 4$.

1.2. Constraints:

1.2.1. $n == \text{colors.length}$



- 1.2.2. $2 \leq n \leq 100$
- 1.2.3. $0 \leq \text{colors}[i] \leq 100$
- 1.2.4. Test data are generated such that at least two houses have different colors.

2. Draw the below pattern with dynamic input rows (the below example is when the input is 5):

```
/tmp/DXSZDRP8kK.o
Enter the number of rows: 5
* * * * *
 *           *
  *         *
   *       *
    *     *
     *   *
      * *
       *
        *
```

3. Maximum Difference by Remapping a Digit

You are given a positive integer `num`. You know that John will sneakily remap exactly one digit from one of the 10 possible digits (0 to 9) to another digit.

Return the difference between the maximum and minimum values John can make by remapping exactly one digit in `num`.

- 3.1. Notes:

- 3.1.1. John can remap different digits to obtain the minimum and maximum values, respectively.

3.1.1.1. First, he will create the maximum value from a given number

3.1.1.2. Then, he will create the minimum value from a given number.

- 3.1.2. Your task is to compute the difference between the maximum and minimum numbers that John can generate by such a remapping. So the function should return the difference between the maximum and minimum values John can obtain.

- 3.1.3. When John remaps a digit `d1` to another digit `d2`, John replaces all occurrences of `d1` in `num` with `d2`. Means if '9' replaces '4', all the '4' occurrences in the number will be replaced by 4. E.g., 4845 -> 9895.



3.1.4. John can remap a digit to itself (means 9 can be replaced/remapped by 9 as well), in which case `num` does not change.

3.1.5. The resulting number after remapping can contain leading zeroes.

3.2. Examples:

Input: num = 11891

Output: 99009

Explanation:

To achieve the maximum value, John can remap the digit 1 to the digit 9, yielding 99899. Remap $1 \rightarrow 9$, $11891 \rightarrow 99899$

To achieve the minimum value, John can remap the digit 1 to the digit 0, yielding 890. Remap $1 \rightarrow 0$, $11891 \rightarrow 00890 \rightarrow$ treated as 890

The difference between these two numbers ($99899 - 890$) is 99009.

Input: num = 90

Output: 99

Explanation:

The maximum value that can be returned by the function is 99 (if 0 is replaced by 9) (Remap: $0 \rightarrow 9$, $90 \rightarrow 99$), and the minimum value that can be returned by the function is 0 (if 9 is replaced by 0) (Remap: $9 \rightarrow 0$, $90 \rightarrow 00 \rightarrow$ treated as 0). Thus, it will return $(99 - 0) 99$.

Input: num = 456

Output: 900

Explanation:

The maximum value that can be returned by the function is 956 (if 4 is replaced by 9) (Remap: $4 \rightarrow 9$, $456 \rightarrow 956$), and the minimum value that can be returned by the function is 56 (if 4 is replaced by 0) (Remap: $4 \rightarrow 0$, $456 \rightarrow 056 \rightarrow$ treated as 56). Thus, it will return $(956 - 56) 900$.

3.3. Constraints:

3.3.1. $1 \leq \text{num} \leq 10^8$



Instructions (Read Carefully): -

1. Time Duration: **3 hours.**
2. Write Programs in a dynamic and interactive way and use OOPs wherever necessary.
3. All the inputs must be taken from the User and it should be dynamic. Do not use static input/values.
4. Do not use any built-in functions/methods in any of the programs. Create your own functions/methods.
5. You do not have permission to access Google.
6. You don't have permission to access Social Networking sites.
7. You can't download anything and can't copy-paste code.
8. you can not use any AI chatbot or code generating tools like chatgpt, gemini,
9. you can not use browsers AI mode
10. you can not use any websites chatbot for any questions
11. Keep your phone on silent and in your bag.
12. A few guidelines to make your code acceptable and JavaScript special instructions: [☰ Interview Guidelines](#)